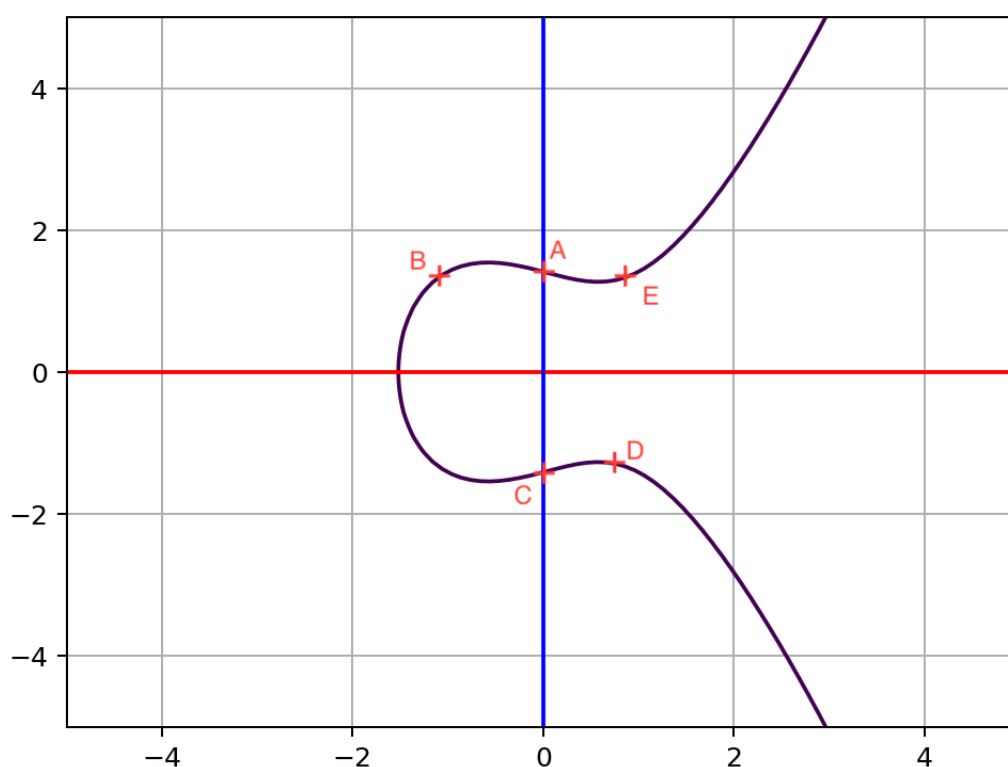


Family Name	First Name
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Part 1 of 3 : Elliptic Curves (2 marks)

Given the following curve

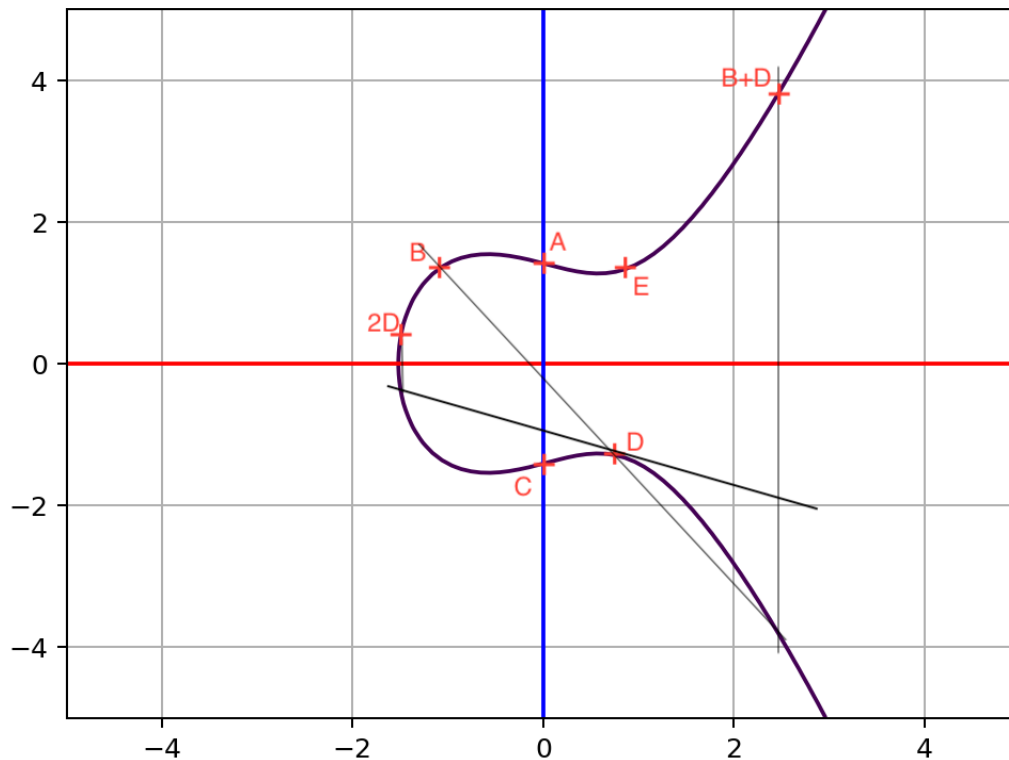


Calculate the following (you can provide the result by both, drawing or specifying the point of the curve if it is already defined):

- a) $A+C$
- b) $B+E$
- c) $2D$
- d) $B+D$

Solution

- a) $A+C = 0$, no point in figure
- b) $B+E = C$
- c) $2D \rightarrow$ Figure
- d) $B+D \rightarrow$ Figure



Part 2 of 3: Diffie-Hellman in Elliptic Curves (3 marks)

Alice and Bob want to agree on a shared secret key using a Diffie-Hellman key exchange protocol based on Elliptic Curves. They pick the curve $E_{11}(1,0)$ and the generator $G=(7,3)$, This is $p=11$, $a=1$ and $b=0$. Using the table of point additions below answer the following questions:

- a) If Alice picks her secret number $n_a=4$, calculate which values does she send to Bob (0,5 mark)
- b) If Bob picks $n_b=5$ as his secret number, calculate which values does he send to Alice (0,5 mark)
- c) If an attacker Charlie, who has access to communication channel between Alice and Bob is able to obtain the values they exchange and is also able to obtain Bob's secret number (via hacking for example), will he be able to obtain the shared key between them? Justify your answer, and if the answer is yes, calculate the final shared value from the point of view of Charlie. (2 marks)

+	∞	(0,0)	(5,3)	(5,8)	(7,3)	(7,8)	(8,5)	(8,6)	(9,1)	(9,10)	(10,3)	(10,8)
∞	∞	(0,0)	(5,3)	(5,8)	(7,3)	(7,8)	(8,5)	(8,6)	(9,1)	(9,10)	(10,3)	(10,8)
(0,0)	(0,0)	∞	(9,10)	(9,1)	(8,6)	(8,5)	(7,8)	(7,3)	(5,8)	(5,3)	(10,8)	(10,3)
(5,3)	(5,3)	(9,10)	(5,8)	∞	(10,8)	(8,6)	(7,3)	(10,3)	(0,0)	(9,1)	(7,8)	(8,5)
(5,8)	(5,8)	(9,1)	∞	(5,3)	(8,5)	(10,3)	(10,8)	(7,8)	(9,10)	(0,0)	(8,6)	(7,3)
(7,3)	(7,3)	(8,6)	(10,8)	(8,5)	(9,10)	∞	(0,0)	(5,3)	(7,8)	(10,3)	(5,8)	(9,1)
(7,8)	(7,8)	(8,5)	(8,6)	(10,3)	∞	(9,1)	(5,8)	(0,0)	(10,8)	(7,3)	(9,10)	(5,3)
(8,5)	(8,5)	(7,8)	(7,3)	(10,8)	(0,0)	(5,8)	(9,1)	∞	(10,3)	(8,6)	(5,3)	(9,10)
(8,6)	(8,6)	(7,3)	(10,3)	(7,8)	(5,3)	(0,0)	∞	(9,10)	(8,5)	(10,8)	(9,1)	(5,8)
(9,1)	(9,1)	(5,8)	(0,0)	(9,10)	(7,8)	(10,8)	(10,3)	(8,5)	(5,3)	∞	(7,3)	(8,6)
(9,10)	(9,10)	(5,3)	(9,1)	(0,0)	(10,3)	(7,3)	(8,6)	(10,8)	∞	(5,8)	(8,5)	(7,8)
(10,3)	(10,3)	(10,8)	(7,8)	(8,6)	(5,8)	(9,10)	(5,3)	(9,1)	(7,3)	(8,5)	(0,0)	∞
(10,8)	(10,8)	(10,3)	(8,5)	(7,3)	(9,1)	(5,3)	(9,10)	(5,8)	(8,6)	(7,8)	∞	(0,0)

Figure 1: Table of Point Additions for the Curve $E_{11}(1,0)$

Solutions

- Alice will send (5,8) to Bob: $4xG$.
- Bob will send (8,5) to Alice: $5xG$.
- If Charlie is able to hack Bob and gets his secret number, he will be able to obtain the shared secret exactly in the same way as Bob would (1 mark). In order to do so he will have to do $5x(5,8)$ which is equals to (5,3) (another mark for the calculations)

Part 3 of 3: Hash Functions (5 marks)

Answer the following questions:

- Describe the one-way property. Provide an example of a hash function that doesn't have this property and a function that has it (2 marks)
- Explain how an attacker would be able to recover a password from a stolen password database if the database only holds the hashed password with SHA-256, $H_{SHA-256}(\text{password})$. Explain what could be done to make this attack more difficult (3 marks)

Solutions:

- The one-way property states that given a hash value $H(m)$ it must be computationally impossible to obtain the m that generated that hash. An example of hash function that doesn't have this property is one that only considers the last n bits of the message. Functions such as SHA-2 have this property.
- The attacker can create a very large list of possible passwords by using common passwords, combination of words, dates, etc. Then they can pre-compute the SHA-256 value of all those passwords and compare the result with the hashed passwords they stole. If there is a match, they have found one of the passwords. It is possible to make this attack more difficult by concatenating a salt to the password before hashing. In this way, the attacker won't be able

to use pre-computed values and will need to compute them after they obtain access to the passwords.